

2026-06-03-phase2-flatten-migration-design

Phase 2 — Flatten own-org submodules to submodules/<leaf> + rewrite refs (Design / Migration Map)

Date: 2026-06-03 **Status:** Approved (operator decisions 2026-06-03) — execution authorized, subagent-driven, incremental with build verification. **Risk:** VERY HIGH / hard-to-reverse. Restore anchor: docs/migration/phase0-restore-manifest-2026-06-03.txt.

Operator decisions (2026-06-03)

1. **Drop** the 4 unused vasic-digital forks (doc_processor, llm_orchestrator, llm_provider, vision_engine) — build consumes the HelixDevelopment forks. Also drop the duplicate **HD** models mount (true dup of vd/models). Upstream repos remain intact; only the submodule mounts are removed.
2. **Keep** constitution at ./constitution/ — exempt from flattening (governance root reached by path-relative @import + name-hardcoded resolver). No constitution reference rewrite needed.

Target layout

Every own-org submodule EXCEPT constitution (and except third-party, which stay nested) lives at submodules/<snake_case_leaf>. Inner Go app helix_code/ stays put (it is a tracked subdir, NOT a submodule).

DROP (5 mounts — git submodule deinit -f + git rm + delete .git/modules/<path> + remove .gitmodules stanza)

- submodules/models (dup of vd/models)
- submodules/doc_processor
- submodules/llm_orchestrator
- submodules/llm_provider
- submodules/vision_engine

MOVE → submodules/<leaf> (65 total)

- **Top-level (8):** challenges, containers, security, panoptic, github_pages_website, helix_agent, helix_qa, mcp_servers
- **HelixDevelopment kept (9):** doc_processor, llm_orchestrator, llm_provider, vision_engine, llms_verifier, helix_memory, helix_specifier, helix_llm, debate_orchestrator
- **vasic-digital kept (48):** models + the 47 others (agentic, auth, auto_temp, background_tasks, benchmark, cache, claritas, concurrency, config, conversation, database, document, embeddings, event_bus, filesystem, formatters, gandalf_solutions, hyper_tune, i18n, i_llm, lazy, leak_hub, llm_ops, mcp_module, memory, messaging, middleware, normalize, observability, optimization, ouroboros, planning, plinius_common, plugins,

rag, rate_limiter, recovery, red_team, self_improve, skill_registry, storage, streaming,
tool_schema, toon, vector_db, veritas, watcher)

Canonical winners for the 5 collision leaves: submodules/models ← vd; submodules/
{doc_processor, llm_orchestrator, llm_provider, vision_engine} ← HD.

go.mod replace rewrites (62 lines / build-critical) — rule by consumer location

Relative replace targets recompute by where the CONSUMING go.mod sits after the move: - helix_code/go.mod (stays at helix_code/, depth-1): ../dependencies/{org}/X → ../submodules/X. (6 lines: docprocessor, llmorchestrator, visionengine, debate→debate_orchestrator, helixspecifier, lazy.) - helix_agent/go.mod (moves to submodules/helix_agent/): siblings under submodules/ → ../dependencies/{org}/X → ../X; ../submodules/llms_verifier/llm-verifier → ../llms_verifier/llm-verifier. (37 lines.) - helix_qa/go.mod (moves to submodules/helix_qa/): its existing sibling-form ../X block already resolves correctly once both it and X are siblings under submodules/. Normalize the dual capital/lowercase blocks to ONE lowercase block; fix the LLMsVerifier→llms_verifier casing mismatch (line 130). Net: ../Challenges→../challenges, ../DocProcessor→../doc_processor, etc.; ../submodules/llms_verifier/llm-verifier→../llms_verifier/llm-verifier. - submodules/llms_verifier/go.mod (moves to submodules/llms_verifier/): ../challenges → ../challenges (still sibling — OK; verify). - **Archived copies under docs/improvements/**/** helixcode_sources/go.mod — not part of active build; leave (or update for tidiness, L-risk).

Other reference rewrites

- .gitmodules: 70 own-org path=/url stanzas → git mv updates them; the 5 drops removed; constitution stanza UNCHANGED.
- .codegraph/config.json: own-org paths now under submodules/ must stay in-index per §11.4.79 (they're in-scope by absence of exclusion; verify globs still cover submodules/). Re-index after.
- **Shell scripts (541 files / 1217 lines) + configs (122/313)** referencing dependencies/{org}/X or top-level own-org dirs → rewrite to submodules/X. Highest-volume but mechanical; do via scripted sed with an explicit old→new map, then grep-verify zero stale refs.
- **Markdown (1144/12296)** → rewrite for reference integrity (CONST-052 treats post-rename drift as a violation); regenerate .html/.pdf siblings if exporter present.

Execution order (incremental, build-verified BEFORE commit)

1. mkdir submodules/. Relocate the 65 via git mv <old> submodules/<leaf> (git updates .gitmodules + .git/config). Drop the 5 mounts.
2. Rewrite the 62 go.mod replaces per rules above; fix helix_qa casing/dup-block.
3. Rewrite scripts/configs/.codegraph via scripted old→new map; grep-verify no stale dependencies/vasic-digital/dependencies/HelixDevelopment own-org refs remain (excluding third-party + archived).
4. Rewrite markdown refs; regenerate exports.

5. **VERIFY (gate before any commit):** `cd helix_code && go build ./... && go vet ./...`; same for `helix_agent`, `helix_qa` where buildable. Reference-integrity `grep = 0` stale. Only on green → commit.
6. Commit meta-repo (path moves + `.gitmodules` + `.codegraph` + `scripts/docs`) + each changed submodule (`helix_agent/helix_qa go.mod`) → push all to all upstreams.
7. CONTINUATION close-out; update `docs/improvements/submodule_owned.txt` paths; HXC-001 (CONST-052 rename) closure.

Rollback

Restore manifest pins every pre-move SHA + remote. A broken build at step 5 → `git restore/git checkout -- . + git submodule update` before any commit; nothing is pushed until build is green.